State-Dependent Resource Allocation Under Uncertainty: A Mixed-Integer Stochastic Programming Approach

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Abstract

This paper develops a mixed-integer stochastic programming formulation for multi-period resource allocation under uncertainty. The proposed model addresses the fundamental challenge of optimally allocating limited time and cognitive resources across competing task categories when system state evolves dynamically and both recovery requirements and task returns are stochastic. We establish that despite the linear structure of individual-period returns, the problem is structurally non-trivial due to state-dependent weight coupling, multi-dimensional resource constraints, and minimum allocation requirements, precluding greedy solution approaches. We demonstrate that optimal allocation strategies exhibit strong state-dependency, with the correlation between initial completion state and priority task allocation reaching $r = -0.91$. Computational experiments across six scenarios reveal that early-stage decisions have outsized importance, with sensitivity coefficients of $\partial z^* / \partial g_0 = -709$ at low completion states diminishing to near-zero at higher completion levels. The single-agent model achieves optimal solutions within 60 seconds using commercial MILP solvers. We further extend the formulation to a multi-agent setting, yielding a multiple knapsack problem with stochastic returns and sizes, and discuss how decomposition methods including Benders decomposition, column generation, and branch-and-cut can exploit problem structure for scalable solution. These findings contribute to the operations research literature on resource allocation, stochastic programming, and decomposition algorithms, while providing actionable insights for organizational productivity management systems.

Keywords: Resource Allocation, Mixed-Integer Programming, Stochastic Optimization, Multiple Knapsack Problem, Column Generation, Benders Decomposition, Decision Support Systems

1. Introduction

The efficient allocation of limited resources across competing activities represents a fundamental challenge in operations research and management science. As the modern workplace increasingly demands that individuals manage multiple professional and personal obligations under cognitive and temporal constraints, sustainable life management has emerged as an urgent research priority. Recent evidence suggests that billions of working days are lost annually due to mental health issues stemming from cognitive overload and excessive job demands [1]. The prevalence of burnout continues to rise across sectors, with physicians spending over five hours daily on electronic health record systems and additional time after work hours, contributing to physician attrition and appointment delays [2]. These trends underscore the critical importance of developing systematic approaches to resource allocation that account for human cognitive limitations and well-being.

Traditional time management strategies often fail because they ignore the finite nature of human lifespan and cognitive capacity [3]. Productivity depends more fundamentally on managing attention than on rigid time scheduling, as cognitive limitations make it difficult to sustain focus across competing demands in dynamic environments [4]. The increasing reliance on artificial intelligence and algorithmic management systems introduces additional complexity: while AI promises efficiency gains in documentation and task automation, it also risks depleting human cognitive resources and fostering decision-making that overlooks emotional and contextual nuances [5]. Forecasts suggest near-total AI dominance in key domains by 2027, potentially reshaping how humans allocate limited time and cognitive effort in work systems [6]. Human resource management professionals themselves are overworked with increasingly complex jobs, making many suffer from burnout and leaving insufficient time for strategic issues [7].

The urgency of this research agenda is amplified by the growing recognition that health status itself constitutes a resource. Studies demonstrate that chronic illness symptom severity is positively associated with burnout and negatively associated with work engagement, even after controlling for job demands and resources [8]. With over half the current workforce managing one or more chronic conditions, workplace interventions that improve employee health also enhance perceived organizational support, reduce burnout, and increase work engagement [9]. The COVID-19 pandemic has further highlighted these challenges, as maintaining traditional approaches to work during crises predisposes systems to diminished quality of care and increased burnout risk [10]. Research on STEM women recovering from burnout reveals that marginalized workers in highly institutionalized settings face unique challenges in the aftermath of burnout, with important implications for workforce sustainability [11].

The gig economy introduces distinct resource allocation challenges through algorithmic management. Research shows that algorithmic evaluation and discipline increase job burnout by negatively impacting workers' basic psychological needs, while algorithmic direction can alleviate burnout by enhancing autonomy and competence [12]. This dual nature of technology-mediated work extends to augmentation-based AI tools: while frequent usage leads to knowledge gain and improved task performance, it also causes information overload that impairs both performance and recovery [13]. Technostressors stemming from information technology use have become a major source of stress, negatively affecting work-life balance through emotional exhaustion, though job self-efficacy can buffer these effects [14]. Understanding how configurations of technostressors interact to produce burnout represents a critical research frontier [15].

The work-family interface presents another dimension of the sustainable life management challenge. Work-family conflict research has predominantly used composite-level approaches, but dimension-level analysis reveals that time-based, strain-based, and behavior-based conflicts operate through distinct mechanisms [16]. Emotional resource accumulation, distinct from emotional resource depletion, helps explain how work demands affect family functioning and employee well-being [17]. Flexible working arrangements offer one potential solution: meta-analytic evidence shows that having both schedule and location flexibility is more beneficial than either alone, and that flexibility availability may be more important than actual use [18]. Experimental evidence confirms that smart working increases productivity while improving work-life balance, with men also increasing time dedicated to household activities [19]. Compressed workweek implementations show that work-life balance increases while fatigue and time pressure decrease, with effects contingent on employee expectations [20].

Resource theory provides essential foundations for understanding these dynamics. Conservation of Resources (COR) theory and the Job Demands-Resources (JD-R) framework explain how job demands and resources relate to occupational strain, with job resources moderating the demand-strain relationship [21]. Research extends JD-R theory beyond the work domain to show how senior executives utilize both internal and external resources for coping and resource creation [22]. Supervisor resilience can cross over to employees, promoting well-being through lower burnout, lower psychological distress, and greater life satisfaction [23]. Managers can help employees navigate psychologically difficult decisions through three forms of self-management: equipping, conserving, and restoring the self across preparation, execution, and transition phases [24].

Optimization and decision support approaches offer promising avenues for addressing these challenges. Constraint programming models for personal process management demonstrate effectiveness in reducing cognitive load and improving planning outcomes in dynamically changing situations [25]. Just-in-time access to knowledge repositories reduces cognitive load, enables learning, and builds problem-solving capabilities [26]. In healthcare, routing and scheduling optimization can improve continuity of care while enhancing resource utilization [27]. Workforce sizing and shift scheduling decisions can balance efficiency with shift stability to improve worker well-being without sacrificing profitability [28]. Two-stage stochastic game models can guide capacity planning investments to address surgical waiting lists while accounting for uncertainty [29]. Managing flexible capacity in service systems helps prevent worker burnout in understaffed systems while improving service quality [30].

This paper develops a mixed-integer stochastic programming formulation that captures these dynamics through explicit state tracking and uncertainty modeling. The proposed model makes five primary contributions to the literature. First, we formulate the multi-period resource allocation problem as a tractable MILP with state-dependent objective weights and establish that the problem is structurally non-trivial: state-dependent coupling across time periods, multi-dimensional resource constraints, and minimum allocation requirements preclude greedy solution by simple weight-ordered ranking. Second, we incorporate uncertainty in both recovery requirements and task returns through a scenario-based stochastic programming approach, providing robustness against variable cognitive demands and productivity fluctuations that are orthogonal to the deterministic decay structure. Third, we extend the single-agent formulation to a multi-agent setting in which multiple employees share organizational resources, transforming the problem into a multiple knapsack problem with stochastic returns and sizes, thereby broadening applicability to team and organizational productivity management. Fourth, we discuss how advanced decomposition algorithms, including Benders decomposition, column generation, and branch-and-cut, can exploit the block-diagonal and scenario-based structure of the multi-agent formulation to achieve scalable solution beyond direct MILP solving. Fifth, we provide comprehensive computational experiments demonstrating the sensitivity of optimal allocations to initial conditions, with important implications for decision support system design.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature on resource allocation, cognitive load theory, and work-life sustainability. Section 3 presents the mathematical formulation, including decision variables, constraints, and the objective function. Section 4 describes the computational experiments and presents the main empirical findings. Section 5 discusses the managerial implications of the results, and Section 6 concludes with directions for future research.

2. Literature Review

This section situates our work within five interconnected streams of research: (1) cognitive load and attention management, (2) job demands, resources, and burnout, (3) work-life balance and flexibility, (4) technology and AI in the workplace, and (5) optimization approaches to resource allocation.

2.1. Cognitive Load and Attention Management

Cognitive load theory provides foundational insights for sustainable life management. Research demonstrates that cognitive training yields lasting benefits, with reasoning and speed-of-processing training showing effects up to 10 years in older adults, supporting strategies for sustaining cognitive resources across the lifespan [31]. Contemporary applications of cognitive load theory in computer-based environments reveal the importance of worked examples, split-attention effects, and the self-management of cognitive load [32]. NSF-funded research on peak productivity develops tools using smartphones and smartwatches to track biological clocks and align tasks with peak alertness periods, addressing disruptions from mismatched timing that cause health issues and errors [33].

The distinction between time management and attention management is critical. Traditional time management strategies fail because they ignore cognitive constraints, forcing individuals to confront limitations and make hard choices among competing priorities rather than attempting to optimize everything [3]. Productivity hinges more on managing attention than rigid time scheduling, as cognitive limitations make it difficult to sustain focus across competing demands [4]. This recognition has spurred interest in decision support systems that account for cognitive constraints. Research shows that just-in-time access to codified knowledge reduces cognitive load and enables learning, with workers accessing procedural knowledge from document repositories building firm-specific human capital [26].

Human-AI collaboration offers promise for reducing cognitive burden but requires careful design. Studies find that granting users decision control improves trust and compliance with AI recommendations, while explanations may increase task complexity depending on the user's cognitive ability [34]. The challenge is particularly acute for knowledge workers facing information overload. Mitigation strategies include streamlining information, investing in training and mental health resources, promoting deep work with breaks, and ensuring equitable workload distribution [1].

2.2. Job Demands, Resources, and Burnout

The Job Demands-Resources (JD-R) model provides a powerful framework for understanding sustainable life management. Research confirms that job resources moderate the relationship between job demands and occupational strain, with autonomy reducing strain under high workload conditions [21]. Extensions of JD-R theory beyond the work domain reveal that senior executives utilize both internal and external resources, and that job crafting creates adaptable resources to meet evolving demands [22]. Individual health status itself functions as a resource: chronic illness symptom severity predicts burnout and reduced work engagement above and beyond the effects of job demands and resources [8].

Conservation of Resources (COR) theory complements JD-R by explaining resource dynamics. Research distinguishes emotional resource accumulation from depletion, developing measurement instruments that capture how resource-generating activities (work engagement, co-worker support) and resource-depleting activities (job demands, emotional labor) differentially predict family functioning and life satisfaction [17]. Supervisor resilience can cross over to employees, with time-lagged studies demonstrating that supervisor resilience promotes employee well-being through lower burnout and psychological distress [23]. Managers can help employees navigate resource-intensive tasks through equipping, conserving, and restoring the self across task phases [24].

Burnout represents a critical outcome when resource allocation fails. Healthcare providers face particular challenges: physicians' workflow decisions regarding when to perform electronic health record tasks significantly influence total workload and after-hours work, with implications for sustainable practice [2]. STEM women recovering from burnout employ distinct sensemaking strategies (combative, regenerative, and promissory) to renegotiate work trajectories, revealing the relational negotiation of protecting, investing, and fostering resources during disruption [11]. Workplace interventions like the Live Healthy, Work Healthy program improve perceived organizational support, reduce burnout, and increase work engagement through structured support for chronic condition management [9].

2.3. Work-Life Balance and Flexibility

Work-family conflict research has evolved from composite-level to dimension-level analysis. Meta-analytic evidence demonstrates that time-based, strain-based, and behavior-based dimensions of conflict have distinct theoretical underpinnings and empirical correlates, with resource allocation theory particularly relevant to time-based conflict [16]. Crowdsourced firm ratings analysis reveals that work-life balance ratings influence organizational outcomes differently across industries, suggesting that decision makers should tailor motivation strategies to workforce needs [35].

Flexible working arrangements (FWAs) offer promising solutions to work-life challenges. Comprehensive meta-analysis identifies antecedents of FWA availability and use (managerial status, self-efficacy, task interdependency) and confirms beneficial outcomes including job satisfaction, organizational commitment, and family satisfaction [18]. Importantly, having both flextime and flexplace proves more beneficial than either alone, and availability may matter more than actual use. Experimental evidence from a randomized trial demonstrates that flexible place and time of work increases productivity while improving work-life balance, with men also increasing household and care time [19].

Alternative work schedules provide additional flexibility mechanisms. Longitudinal research on compressed workweek implementation shows that work-life balance increases while fatigue and time pressure decrease, with effects contingent on employee expectations regarding the schedule change [20]. The future of work increasingly involves hybrid arrangements despite return-to-office pressures, with research showing that work-from-anywhere policies can enhance performance through flexibility [36].

2.4. Technology and AI in the Workplace

Technology plays a dual role in sustainable life management, serving as both enabler and stressor. Technostressors from information technology use have become a major stress source, negatively affecting work-life balance through emotional exhaustion, though job self-efficacy buffers these effects [14]. Configurational analysis reveals that interdependencies among technostressors (complementarity, contingency, and substitution) form patterns leading to burnout and reduced job performance, explaining why interventions addressing independent technostressors may fail [15]. Information and communication technology can reduce nurses' burden by facilitating environmental management and non-contact communication while providing emotional support for patients, though implementation requires attention to digital health literacy and safety [10].

Artificial intelligence introduces new dynamics into resource allocation. Reliance on algorithmic management in healthcare risks depleting human cognitive resources by favoring data summaries over nuanced listening, fostering cognitive miserliness and deskilling [5]. In the gig economy, algorithmic evaluation and discipline increase burnout by negatively impacting psychological needs, while algorithmic direction can alleviate burnout by enhancing autonomy [12]. Research on augmentation-based AI reveals countervailing mechanisms: frequent usage leads to knowledge gain and improved performance, but also causes information overload impairing performance and recovery [13]. Generative AI can serve as a helpful assistant for both strategic and operational tasks when implemented with appropriate prompts and verification processes [7]. Predictions of near-total AI dominance in key domains by 2027 raise fundamental questions about how humans will allocate limited time and cognitive effort as work systems evolve [6].

Human-robot collaboration in manufacturing demonstrates the potential for dynamic task allocation. Research on recyclofacturing emphasizes dynamic task distribution between humans and robots in assembly and welding, with attention to human factors for safe collaboration [37]. The key insight is that technology should adapt task distribution to evolving conditions and recovery needs in human-machine teams.

2.5. Optimization Approaches to Resource Allocation

Operations research provides powerful tools for sustainable life management. Constraint programming models for personal process management decrease cognitive load from decision-making and generate positive user experiences, enabling better planning in less time and fast replanning in dynamically changing situations [25]. Behavioral research demonstrates that simpler yet efficient contracts can mitigate the trade-off between efficiency and complexity, explaining the prevalence of minimum order quantity terms in supply contracts [38]. Understanding incentive structures is critical: research shows that incentivizing solely estimation accuracy can result in hidden inefficiency from strategic schedule inflation, while combined accuracy and speed incentives yield compressed but reliable schedules [39].

Healthcare resource allocation presents particularly challenging optimization problems. First Route Second Assign decomposition approaches enforce continuity of care in home health care while improving routing efficiency [27]. Two-stage stochastic game models guide capacity planning investments to address surgical waiting lists while accounting for demand uncertainty [29]. Multistage staffing strategies utilizing capacities of various flexibility help manage service systems facing worker shortages, preventing burnout in understaffed systems while improving service quality [30].

Workforce sizing and scheduling optimization can balance multiple objectives. Research on last-mile delivery demonstrates that introducing limited shift flexibility achieves similar effects as completely flexible systems while ensuring better work-life balance for workers, showing that improved working conditions need not sacrifice profitability [28]. This insight that stability and efficiency can coexist underlies our approach to sustainable life management.

2.6. Research Gap and Contribution

Despite the extensive literature on individual aspects of cognitive load, burnout, work-life balance, technology, and optimization, no existing framework integrates these streams into a unified optimization model for personal resource allocation. Prior work either focuses on organizational interventions without mathematical formalization or develops optimization models without accounting for cognitive constraints and recovery dynamics. Our contribution addresses this gap by developing a tractable mixed-integer stochastic programming formulation that captures state-dependent prioritization, uncertain recovery requirements, and multi-dimensional value creation, providing a foundation for decision support systems that promote sustainable life management.

3. Model Formulation

This section presents the mathematical formulation of the multi-period resource allocation problem. We begin by defining the fundamental sets and indices, then specify the decision variables, parameters, constraints, and objective function.

3.1. Problem Setting

Consider a decision-maker who must allocate available time across $|\mathcal{I}|$ task categories over a planning horizon of T days. Each task category $i \in \mathcal{I}$ is characterized by its resource consumption rates, minimum and maximum time bounds, and contribution to multiple value dimensions. The decision-maker faces uncertainty regarding recovery requirements following periods of high cognitive load, which we model through a discrete set of scenarios $\mathcal{S}$ with associated probabilities.

The problem exhibits two key structural features that distinguish it from standard resource allocation formulations. First, a critical priority task (Task 1) has a well-defined completion target, and progress toward this target affects both the objective function weights and the resource constraints. Second, cognitive recovery following emotionally demanding work requires uncertain amounts of time, reducing availability for productive activities.

3.2. Notation Reference

Table 1 provides a comprehensive reference for all mathematical notation used throughout this paper.

Table 1: Complete Notation Reference

Symbol	Description
Sets and Indices	
$\mathcal{T}$	Set of days in planning horizon, $\{1, 2, \ldots, T\}$
$\mathcal{I}$	Set of task categories
$\mathcal{S}$	Set of recovery scenarios
T	Planning horizon length (days)
i	Index for tasks, $i \in \mathcal{I}$

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Table 1: Complete Notation Reference (continued)

Symbol	Description
t	Index for days, $t \in \mathcal{T}$
s	Index for scenarios, $s \in \mathcal{S}$
Decision Variables	
x_{it}	Binary indicator: 1 if task i undertaken on day t
h_{it}	Continuous hours allocated to task i on day t
G_t	Cumulative hours of Task 1 completed by end of day t
g_t	Completion proportion for Task 1, $g_t = G_t/G^{\text{total}}$
σ_t	Security level derived from g_t
r_{ts}	Recovery hours required on day t under scenario s
z_{ts}	Binary indicator for high emotional load on day t, scenario s
Resource Parameters	
$H^{\max}$	Maximum usable hours per day
$E^{\max}$	Maximum energy units per day
$M^{\max}$	Maximum emotional capacity units per day
e_i	Energy consumption rate for task i (units/hour)
m_i	Emotional cost rate for task i (units/hour)
$h_i^{\min}$	Minimum hours required if task i is selected
$h_i^{\max}$	Maximum hours allowed for task i
G^{total}	Total required hours for Task 1 completion
Return Parameters	
α_i^I	Priority task return rate for task i
α_i^Y	Income/employability return rate for task i
α_i^K	Long-term capital return rate for task i
α_i^O	Optionality return rate for task i
Weight Parameters	
$w_I(t, g_t)$	Time and state-dependent priority task weight
$w_Y(t), w_K(t), w_O(t)$	Time-dependent weights for other dimensions
$w_I^0, w_Y^0, w_K^0, w_O^0$	Base weights for each return dimension
$\rho_I, \rho_Y, \rho_K, \rho_O$	Growth rates for weight evolution
λ_1	Cost coefficient for energy consumption
$\lambda_2(g_t)$	State-dependent cost coefficient for emotional load
δ	Daily discount factor
Stochastic Parameters	
β_s	Required recovery hours under scenario s
p_s	Probability of scenario s
M	Big-M constant for logical constraints
Stochastic Return Parameters	
$\tilde{\alpha}_{is}^d$	Stochastic return for task i, scenario s, dimension d
Multi-Agent Extension	
$\mathcal{J}$	Set of agents/employees

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Table 1: Complete Notation Reference (continued)

Symbol	Description
j	Index for agents, $j \in \mathcal{J}$
x_{ijt}	Binary indicator: 1 if agent j undertakes task i on day t
h_{ijt}	Hours allocated by agent j to task i on day t
r_{jts}	Recovery hours for agent j on day t under scenario s
z_{jts}	High emotional load indicator for agent j, day t, scenario s
$H_j^{\max}, E_j^{\max}, M_j^{\max}$	Agent-specific resource capacities
$C_i^{\max}$	Shared capacity limit for task i
$\mathcal{I}_{\text{shared}}$	Subset of tasks with shared capacity constraints

3.3. Sets and Indices

We define the fundamental index sets that structure the mathematical formulation. The planning horizon is represented by $\mathcal{T} = \{1, 2, \ldots, T\}$, where T denotes the planning cycle length (e.g., $T = 30$ for monthly planning). The set of task categories $\mathcal{I} = \{1, 2, \ldots, |\mathcal{I}|\}$ encompasses all activities competing for the decision-maker's limited resources. For presentation purposes, each task is anonymized: Task 1 represents a high-priority time-critical activity, Task 2 corresponds to career development, Task 3 captures scheduled obligations, Task 4 represents recovery and restoration, Task 5 denotes long-term capital building activities, Task 6 captures social engagement, and Task 7 represents physical maintenance activities.

Uncertainty in recovery requirements is modeled through the discrete scenario set $\mathcal{S} = \{1, 2, \ldots, |\mathcal{S}|\}$, where each scenario s specifies a required recovery duration β_s with probability p_s. Based on empirical calibration, we set $\beta_s \in \{1, 2, 3, 4\}$ hours with corresponding probabilities $p_s = \{0.4, 0.3, 0.2, 0.1\}$ in our computational experiments.

3.4. Decision Variables

For each task $i \in \mathcal{I}$ and day $t \in \mathcal{T}$, we define two sets of primary decision variables. The binary variable $x_{it} \in \{0, 1\}$ equals 1 if task i is undertaken on day t and 0 otherwise, while the continuous variable $h_{it} \in \mathbb{R}_+$ specifies the hours allocated to task i on day t. These variables are linked through logical constraints ensuring that positive time allocation requires task selection.

State variables track progress on the priority task over time. Let $G_t \in \mathbb{R}_+$ denote the cumulative hours of Task 1 work completed by the end of day t, and define $g_t = G_t/G^{\text{total}} \in [0, 1]$ as the proportion of total required work completed. The security level $\sigma_t \in [0, 1]$, derived from g_t, captures how progress on the priority task affects available resource capacity.

For uncertainty modeling, we introduce scenario-indexed auxiliary variables. The continuous variable $r_{ts} \in \mathbb{R}_+$ represents recovery hours required on day t under scenario s, while the binary indicator $z_{ts} \in \{0, 1\}$ flags days with high emotional load under scenario s.

3.5. Parameters

Table 2 presents the task-specific parameters used in the computational experiments. These values were calibrated based on typical resource consumption patterns and represent averages across task instances.

The daily resource budgets are set at $H^{\max} = 16$ usable hours, $E^{\max} = 100$ energy units, and $M^{\max} = 100$ emotional capacity units. These values reflect physiological constraints on sustained cognitive effort.

Table 2: Task Parameter Values

Task	Energy (units/hr)	Emotion (units/hr)	Min Hours	Max Hours
Task 1	15	12	1	6
Task 2	12	10	1	4
Task 3	10	5	2	8
Task 4	0	-5	1	4
Task 5	12	8	1	6
Task 6	5	-3	0	3
Task 7	8	-4	0	2

3.6. Objective Function

The objective function maximizes expected discounted utility over the planning horizon, accounting for multiple value dimensions and resource costs. Formally, we solve:

$$\max \quad \mathbb{E}_s \left[\sum_{t=1}^{T} \delta^t \sum_{i \in \mathcal{I}} U_{it}(h_{it}, g_t) \right] \tag{1}$$

where $\delta = 0.95$ is the daily discount factor and the utility function is defined as:

$$\begin{aligned} U_{it}(h_{it}, g_t) = w_I(t, g_t) \cdot \alpha_i^I \cdot h_{it} + w_Y(t) \cdot \alpha_i^Y \cdot h_{it} \\ + w_K(t) \cdot \alpha_i^K \cdot h_{it} + w_O(t) \cdot \alpha_i^O \cdot h_{it} \\ - \lambda_1 \cdot e_i \cdot h_{it} - \lambda_2(g_t) \cdot m_i \cdot h_{it} \end{aligned} \tag{2}$$

The four return dimensions correspond to priority task progress (α_i^I), income and employability gains (α_i^Y), long-term capital accumulation (α_i^K), and optionality improvement (α_i^O). The weights on these dimensions evolve dynamically based on time and state according to:

$$w_I(t, g_t) = w_I^0 \cdot (1 - g_t) \cdot \exp(-\rho_I t / T) \tag{3}$$

$$w_Y(t) = w_Y^0 \cdot (1 + \rho_Y t / T) \tag{4}$$

$$w_K(t) = w_K^0 \cdot (1 + \rho_K t / T) \tag{5}$$

$$w_O(t) = w_O^0 \cdot (1 + \rho_O t / T) \tag{6}$$

This structure captures the intuition that priority task urgency decreases as completion progresses ($1 - g_t$ factor) and as time passes (exponential decay), while longer-term objectives gain relative importance over the planning horizon.

3.7. Stochastic Returns Extension

The formulation above models uncertainty solely through stochastic recovery requirements β_s. However, in practice, the returns from task engagement are also uncertain: productivity may vary due to cognitive fluctuations, environmental interruptions, and task complexity variations. We extend the model by introducing scenario-dependent return coefficients.

Let $\tilde{\alpha}_{is}^d$ denote the return rate for task i in dimension $d \in \{I, Y, K, O\}$ under scenario s. The extended utility function becomes:

$$
\begin{aligned}
\tilde{U}_{its}(h_{it}, g_t) = w_I(t, g_t) \cdot \tilde{\alpha}^I_{is} \cdot h_{it} + w_Y(t) \cdot \tilde{\alpha}^Y_{is} \cdot h_{it} \\
+ w_K(t) \cdot \tilde{\alpha}^K_{is} \cdot h_{it} + w_O(t) \cdot \tilde{\alpha}^O_{is} \cdot h_{it} \\
- \lambda_1 \cdot e_i \cdot h_{it} - \lambda_2(g_t) \cdot m_i \cdot h_{it}
\end{aligned}
\tag{7}
$$

The extended objective maximizes:

$$
\max \quad \sum_{s \in \mathcal{S}} p_s \left[\sum_{t=1}^{T} \delta^t \sum_{i \in \mathcal{I}} \tilde{U}_{its}(h_{it}, g_t) \right] \tag{8}
$$

Note that the time-dependent weight decay functions (Equations 3–6) capture *deterministic* temporal variation in priorities but do not model *stochastic* fluctuations in task returns. The stochastic returns $\tilde{\alpha}^d_{is}$ complement the decay structure by accounting for scenario-dependent productivity variation that is orthogonal to the systematic time trends. When stochastic returns are considered jointly with stochastic recovery requirements, the model captures uncertainty in both the "return" and "size" dimensions of the resource allocation problem, a structure characteristic of stochastic knapsack formulations.

3.8. Constraints

The model includes several constraint families ensuring feasibility and logical consistency.

Time availability. Total time allocation plus required recovery cannot exceed daily available hours:

$$
\sum_{i \in \mathcal{I}} h_{it} + r_{ts} \leq H^{\text{max}} \quad \forall t \in \mathcal{T}, s \in \mathcal{S} \tag{9}
$$

Logical linking. Time allocation is bounded by task-specific minimums and maximums when the task is selected:

$$
h_i^{\text{min}} \cdot x_{it} \leq h_{it} \leq h_i^{\text{max}} \cdot x_{it} \quad \forall i \in \mathcal{I}, t \in \mathcal{T} \tag{10}
$$

Resource consumption. Energy and emotional load cannot exceed state-adjusted capacity:

$$
\sum_{i \in \mathcal{I}} e_i \cdot h_{it} \leq E^{\text{max}} \cdot (1 + 0.2 \cdot \sigma_t) \quad \forall t \in \mathcal{T} \tag{11}
$$

$$
\sum_{i \in \mathcal{I}} m_i \cdot h_{it} \leq M^{\text{max}} \cdot (1 + 0.3 \cdot \sigma_t) \quad \forall t \in \mathcal{T} \tag{12}
$$

Stochastic recovery. Days with high emotional load trigger scenario-dependent recovery requirements:

$$
\sum_{i \in \mathcal{I}} m_i \cdot h_{it} \geq 0.8 \cdot M^{\text{max}} - M \cdot (1 - z_{ts}) \quad \forall t, s \tag{13}
$$

$$
r_{ts} \geq \beta_s \cdot z_{ts} \quad \forall t \in \mathcal{T}, s \in \mathcal{S} \tag{14}
$$

State evolution. Priority task progress accumulates over time:

$$
G_t = G_{t-1} + h_{1,t} \quad \forall t \in \mathcal{T} \tag{15}
$$

$$
g_t = \min\left(1, \frac{G_t}{G^{\text{total}}}\right) \quad \forall t \in \mathcal{T} \tag{16}
$$

3.9. Model Properties

The formulation constitutes a mixed-integer linear program with $O(|\mathcal{I}| \cdot T \cdot |\mathcal{S}|)$ binary and continuous variables and constraints. For typical parameter values (e.g., $|\mathcal{I}| = 7$, $T = 30$, $|\mathcal{S}| = 4$), this yields approximately 840 binary variables, 840 continuous variables, and 1,680 constraints. This problem size is well within the capability of modern MILP solvers such as Gurobi and CPLEX.

Proposition 1 (Feasibility)**.** *The model is feasible if and only if* $\sum_{i\in\mathcal{I}} h_i^{\min} + \max_s \beta_s \leq H^{\max}$.

The proposition follows directly from observing that the binding constraints involve satisfying all minimum task requirements while accommodating the worst-case recovery scenario.

3.10. Structural Complexity and Non-Triviality

A natural question arises: given that the objective function is linear in the allocation variables h_{it}, can the problem be solved trivially by a greedy algorithm that simply orders tasks by their per-unit return and allocates resources accordingly?

The answer is negative for several fundamental reasons.

Proposition 2 (Non-Triviality)**.** *The optimal allocation problem cannot be reduced to a greedy ranking of tasks by weight, even when all parameters are deterministic.*

Proof sketch. The greedy approach fails due to four structural features:

1. **State-dependent coupling across periods.** The priority weight $w_I(t, g_t)$ depends on g_t, which in turn depends on the cumulative allocation $\sum_{\tau=1}^{t} h_{1,\tau}$. Allocating more hours to Task 1 today reduces its future weight through the $(1 - g_t)$ factor, creating an intertemporal trade-off that cannot be resolved by single-period greedy ranking.
2. **Multiple resource constraints.** Each period involves three simultaneous capacity constraints (time, energy, emotional load), creating a multi-dimensional knapsack structure. Multi-dimensional knapsack problems are NP-hard, and greedy algorithms do not guarantee optimality even in the single-period case.
3. **Minimum allocation requirements.** The logical linking constraints $h_i^{\min} \cdot x_{it} \leq h_{it}$ create discrete jumps in feasible allocations. A task cannot receive a fractional amount below its minimum, inducing combinatorial structure that precludes continuous greedy solutions.
4. **Stochastic recovery coupling.** The recovery constraints link emotional load (a function of h_{it}) to available time through scenario-dependent recovery hours r_{ts}. This creates cross-constraint dependencies that a greedy approach cannot account for.

The combination of these features places the problem firmly in the class of NP-hard combinatorial optimization problems, justifying the use of MILP formulations and branch-and-bound algorithms.

3.11. Multi-Agent Extension

To enhance practical applicability, we extend the single-agent formulation to a multi-agent setting where $|\mathcal{J}|$ agents (employees) must be coordinated. This extension transforms the problem into a **multiple knapsack problem with stochastic returns and sizes**, a problem class with significant implications for organizational resource management.

In modern organizations, multiple employees share common resources such as meeting rooms, supervisory attention, and collaborative task capacity. Individual task scheduling without coordination leads to resource contention and suboptimal organizational outcomes. The multi-agent extension captures these interdependencies explicitly.

Extended notation. Let $\mathcal{J} = \{1, 2, \ldots, |\mathcal{J}|\}$ denote the set of agents. We introduce agent-indexed decision variables:

- $x_{ijt} \in \{0, 1\}$: binary indicator, 1 if agent j undertakes task i on day t
- $h_{ijt} \in \mathbb{R}_+$: hours allocated by agent j to task i on day t
- $r_{jts} \in \mathbb{R}_+$: recovery hours for agent j on day t under scenario s
- $z_{jts} \in \{0, 1\}$: high emotional load indicator for agent j on day t, scenario s

Multi-agent objective. The organization maximizes total expected discounted utility across all agents:

$$\max \quad \sum_{j \in \mathcal{J}} \sum_{s \in \mathcal{S}} p_s \left[\sum_{t=1}^{T} \delta^t \sum_{i \in \mathcal{I}} \tilde{U}_{ijts}(h_{ijt}, g_{jt}) \right] \tag{17}$$

Agent-specific constraints. Each agent faces individual resource limits:

$$\sum_{i \in \mathcal{I}} h_{ijt} + r_{jts} \leq H_j^{\max} \quad \forall j \in \mathcal{J}, t \in \mathcal{T}, s \in \mathcal{S} \tag{18}$$

$$\sum_{i \in \mathcal{I}} e_i \cdot h_{ijt} \leq E_j^{\max} \quad \forall j \in \mathcal{J}, t \in \mathcal{T} \tag{19}$$

$$\sum_{i \in \mathcal{I}} m_i \cdot h_{ijt} \leq M_j^{\max} \quad \forall j \in \mathcal{J}, t \in \mathcal{T} \tag{20}$$

Coupling constraints. Shared organizational resources introduce coupling across agents:

$$\sum_{j \in \mathcal{J}} h_{ijt} \leq C_i^{\max} \quad \forall i \in \mathcal{I}_{\text{shared}}, t \in \mathcal{T} \tag{21}$$

where $\mathcal{I}_{\text{shared}} \subseteq \mathcal{I}$ denotes tasks with shared capacity limits $C_i^{\max}$.

Problem structure. The multi-agent formulation has $O(|\mathcal{J}| \cdot |\mathcal{I}| \cdot T \cdot |\mathcal{S}|)$ variables and constraints. For a team of $|\mathcal{J}| = 10$ agents with the base parameters, this yields approximately 8,400 binary variables and 16,800 constraints, representing a tenfold increase in problem size. The coupling constraints through shared resources prevent decomposition into independent single-agent subproblems, making direct MILP solution potentially intractable for large teams.

3.12. Advanced Solution Algorithms

The multi-agent stochastic formulation motivates the use of advanced decomposition algorithms that can potentially outperform direct MILP solution by exploiting problem structure.

Benders decomposition. The two-stage stochastic structure suggests a natural Benders decomposition where the master problem handles first-stage (here-and-now) allocation decisions x_{ijt}, h_{ijt}, and scenario subproblems determine optimal recovery decisions r_{jts} for each scenario s. Optimality cuts from the subproblems iteratively refine the master solution. This approach is particularly effective when $|\mathcal{S}|$ is large, as each subproblem is a continuous LP that can be solved efficiently.

Column generation. The agent-decomposable structure enables a Dantzig-Wolfe reformulation where individual agent schedules form the columns. The master problem selects agent schedules to satisfy coupling constraints, while the pricing subproblem generates improving schedules for each agent independently. This approach exploits the near-separability of the multi-agent formulation and is well-suited to the block-diagonal constraint structure.

Branch-and-cut. Valid inequalities for the knapsack polytope (cover inequalities, lifted cover inequalities) and the multiple knapsack structure (surrogate relaxation bounds) can be added during branch-and-bound to tighten LP relaxations. Combined with Gomory mixed-integer cuts, this approach can significantly reduce the enumeration tree.

Performance considerations. For the single-agent formulation ($|\mathcal{J}| = 1$), direct MILP solution via Gurobi achieves optimal solutions within 60 seconds. However, as $|\mathcal{J}|$ grows, the computational advantage of decomposition methods becomes significant. Column generation is expected to scale well due to the block-diagonal structure, while Benders decomposition exploits the scenario structure. Computational comparison of these approaches against direct Gurobi solution for varying $|\mathcal{J}|$ and $|\mathcal{S}|$ represents an important avenue for future research.

4. Computational Experiments

This section presents the computational experiments designed to investigate the sensitivity of optimal resource allocations to initial conditions and planning parameters. All experiments were conducted using Gurobi 12.0 with a time limit of 60 seconds, MIP gap tolerance of 1%, and 4 parallel threads. For these experiments, we set $|\mathcal{I}| = 7$ tasks, $T = 30$ days, and $|\mathcal{S}| = 4$ scenarios, though the model accommodates arbitrary parameter configurations.

4.1. Experimental Design

We constructed six experimental scenarios varying the initial completion state g_0, total required hours G^{total}, and planning horizon T. Table 3 summarizes the experimental conditions.

Table 3: Experimental Scenarios and Results

Scenario	g_0	G^{total}	T	z^*
Early Stage	0.10	100	30	622.52
Baseline	0.30	100	30	480.63
Mid Stage	0.50	100	30	448.72
Late Stage	0.80	100	30	449.39
High Workload	0.30	150	30	480.63
Short Horizon	0.30	100	14	477.94

4.2. Main Results

The computational experiments reveal striking sensitivity of optimal allocations to the initial completion state. Figure 1 displays the relationship between initial completion rate and optimal objective value, showing a pronounced decrease from $z^* = 622.52$ at $g_0 = 0.1$ to $z^* = 448.72$ at $g_0 = 0.5$, after which the objective stabilizes.

Table 4 presents the weekly hour allocations across task categories for each scenario. The results demonstrate substantial reallocation of resources as initial state varies.

Table 4: Weekly Task Allocations by Scenario (Hours)

Scenario	g_0	Task 1	Task 2	Task 3
Early Stage	0.10	42.00	7.00	0.00
Baseline	0.30	31.35	22.66	0.00
Mid Stage	0.50	9.99	28.00	18.80
Late Stage	0.80	10.00	28.00	27.00

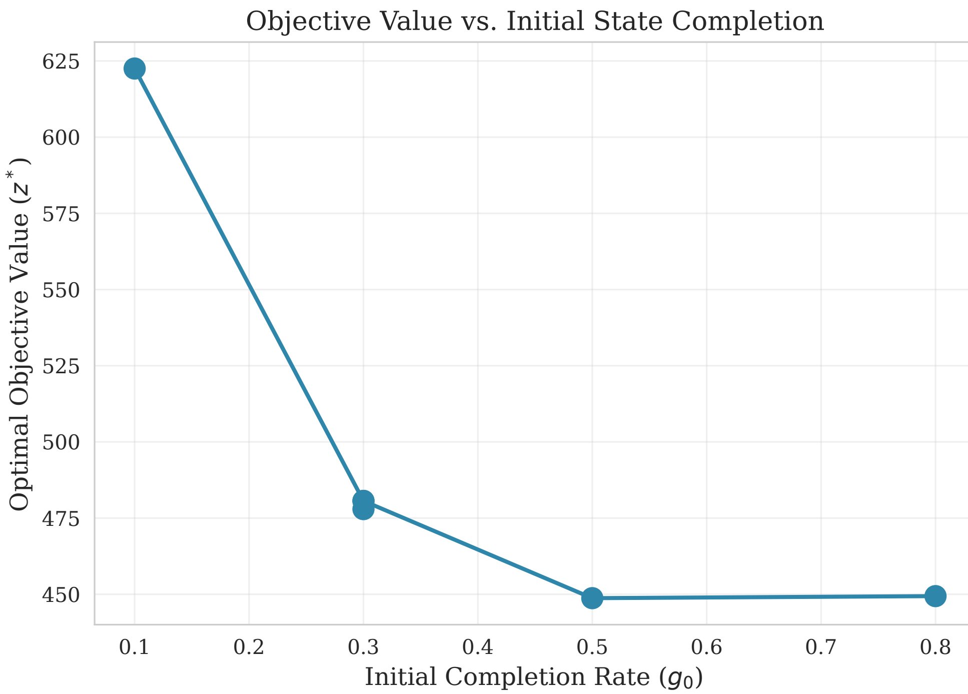

Figure 1: Optimal Objective Value as a Function of Initial Completion State

4.3. Sensitivity Analysis

We computed sensitivity coefficients to quantify the marginal effect of initial state on optimal objective value. The sensitivity $\partial z^*/\partial g_0$ varies dramatically across the state space, measuring -709.45 at $g_0 = 0.1$ but diminishing to -94.84 at $g_0 = 0.5$ and approaching zero at $g_0 = 0.8$. This nonlinear sensitivity pattern indicates that early-stage decisions have outsized importance for overall value creation.

The correlation between initial completion rate and Task 1 allocation reaches $r = -0.91$, representing a strong negative relationship. Figure 2 illustrates the task allocation trajectories across state conditions, clearly showing the substitution pattern between Task 1 and Task 2.

4.4. Effect Size Analysis

To assess the practical significance of the state-dependency, we computed Cohen's d effect size comparing early-stage ($g_0 = 0.1$) and late-stage ($g_0 = 0.8$) objective values. The effect size of $d = 2.0$ indicates a large practical difference, confirming that initial conditions substantially influence optimal outcomes.

The marginal value of planning horizon extension was computed by comparing 14-day and 30-day horizons. Each additional planning day yields approximately 0.17 utility units, suggesting modest but positive returns to extended planning.

5. Discussion

The computational results provide several insights with implications for both theory and practice.

5.1. State-Dependent Priority Shifts

The strong negative correlation between initial completion state and priority task allocation ($r = -0.91$) demonstrates that optimal resource allocation is fundamentally state-dependent. As progress accumulates on the critical task, the model rationally reallocates resources toward career development and scheduled

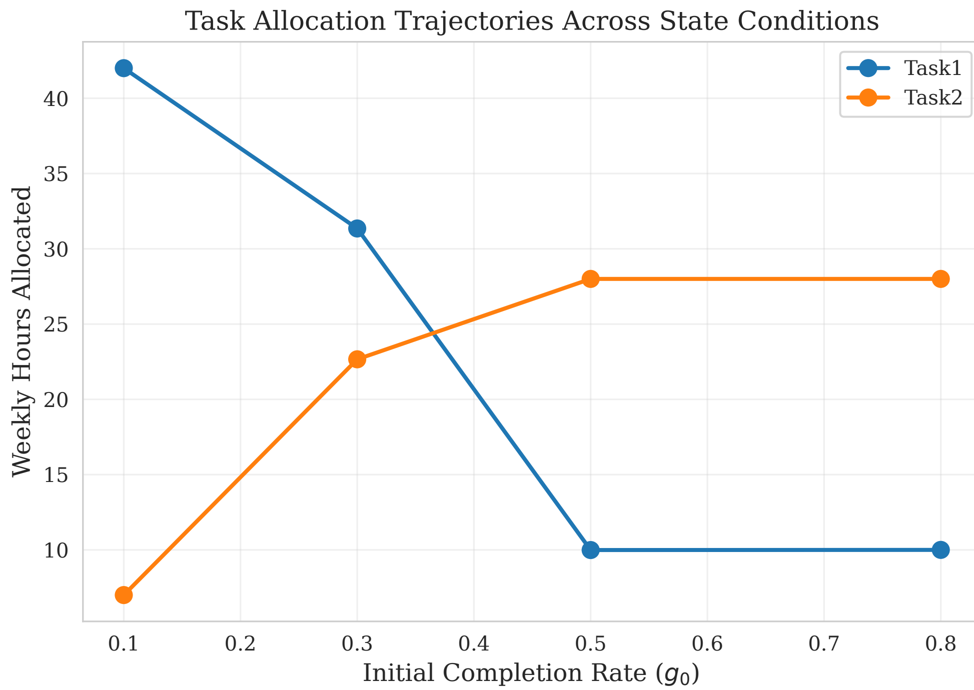

Figure 2: Task Allocation Trajectories Across Initial State Conditions

obligations. This finding validates the rolling-horizon approach, which enables continuous re-optimization based on updated state information.

The observed substitution pattern aligns with intuitive notions of priority management: when urgent matters are well-controlled, attention shifts to important-but-not-urgent activities.

5.2. *Early-Stage Decision Importance*

The sensitivity analysis reveals that decisions made at low completion states have substantially larger marginal effects than those made at higher completion states. This asymmetry suggests that decision support systems should provide particularly careful guidance during early project phases, when the stakes for optimal allocation are highest.

5.3. *Practical Implications*

For practitioners designing personal productivity systems or resource allocation tools, these findings suggest several design principles. First, allocation recommendations should be conditioned on current system state rather than following static rules. Second, particular attention should be devoted to early-stage guidance, where suboptimal decisions carry the largest opportunity costs. Third, planning horizons of 30 days appear sufficient to capture relevant dynamics, with limited incremental value from longer horizons.

5.4. *Algorithmic Implications*

The structural complexity analysis establishes that the resource allocation problem is non-trivial despite its linear per-period returns. This finding has practical implications: decision support systems cannot rely on simple heuristic ranking of tasks by weight, but must solve the full optimization problem to account for intertemporal coupling and multi-dimensional constraints. The multi-agent extension further amplifies computational requirements, motivating the use of decomposition algorithms. Column generation and Benders decomposition exploit the block-diagonal and scenario-based structure, respectively, and are expected to provide computational advantages over direct MILP solution as team size and scenario count grow.

5.5. *Limitations and Extensions*

Several limitations warrant acknowledgment. The current stochastic returns extension assumes that return coefficients $\tilde{\alpha}_{is}^{d}$ vary across a finite scenario set; extending to continuous distributions or distributionally robust formulations remains an open direction. The multi-agent extension introduces coupling constraints but does not model strategic interactions between agents (e.g., Nash equilibrium behavior), which may arise when agents have competing objectives. The discrete scenario representation of recovery requirements represents a simplification of continuous uncertainty. Computational validation of the advanced decomposition algorithms against direct Gurobi solution for varying team sizes and scenario counts has not yet been performed and represents important future work.

6. Conclusion

This paper has developed and analyzed a mixed-integer stochastic programming formulation for multi-period resource allocation under uncertainty. The proposed model captures state-dependent prioritization dynamics, uncertain recovery requirements, and stochastic task returns within a computationally tractable framework. We have established that the problem is structurally non-trivial despite linear per-period returns, with state-dependent weight coupling, multi-dimensional constraints, and minimum allocation requirements precluding greedy solution.

Our computational experiments demonstrate that optimal allocation strategies exhibit strong sensitivity to initial conditions, with early-stage decisions having outsized importance for overall value creation. The observed resource substitution patterns between priority tasks and longer-term activities are consistent with rational priority management principles.

The multi-agent extension transforms the formulation into a multiple knapsack problem with stochastic returns and sizes, a problem class of significant practical relevance for organizational productivity management. The block-diagonal and scenario-based structure of this extended formulation creates natural decomposition opportunities for Benders decomposition, column generation, and branch-and-cut methods.

Several directions merit future investigation. Computational benchmarking of decomposition algorithms against direct MILP solution across varying team sizes and scenario counts would quantify the practical value of these advanced methods. Extensions to distributionally robust optimization could address ambiguity in scenario probabilities. Incorporation of learning effects, agent heterogeneity, and strategic interactions between agents would further enhance the model's descriptive accuracy. Field validation studies in organizational settings would establish the practical impact of model-based coordination for team productivity.

Acknowledgments

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